

Week 7 : Assignment 7

Detailed Solutions

Question 1

Which among the following is not a type of cross-validation technique?

- LOOCV
- k -fold cross validation
- Validation set approach
- ✓ Bias variance trade off

Solution:

Cross-validation is a technique used to evaluate the performance of a machine learning model by dividing the available data into training and validation portions.

Some commonly used validation techniques are:

- **LOOCV (Leave-One-Out Cross-Validation):** One observation is kept for validation and all remaining observations are used for training. This process is repeated for every observation.
- **k -fold cross-validation:** The dataset is divided into k subsets or folds. The model is trained on $k - 1$ folds and tested on the remaining fold. This process is repeated k times.
- **Validation set approach:** The available dataset is divided into a training set and a validation set.

However, **bias-variance trade-off** is not a cross-validation technique. It is a fundamental concept describing the trade-off between model bias and model variance.

Bias variance trade off

Question 2

Which among the following is a classification problem? (*Select all that apply.*)

- ☐ Predicting the average rainfall in a given month.
- ☒ Predicting whether a patient is diagnosed with a disease or not.
- ☐ Predicting the price of a house.
- ☒ Predicting whether it will rain or not tomorrow.

Solution:

A **classification problem** is a machine learning problem where the target variable belongs to a finite set of categories or classes.

For example, if we want to predict whether a patient has a disease:

$$Y = \begin{cases} 1, & \text{Disease diagnosed} \\ 0, & \text{Disease not diagnosed} \end{cases}$$

This is a binary classification problem because there are two possible classes.

Similarly, predicting whether it will rain tomorrow can be represented as:

$$Y = \begin{cases} 1, & \text{It will rain} \\ 0, & \text{It will not rain} \end{cases}$$

Therefore, this is also a binary classification problem.

On the other hand, average rainfall and house price are both numerical quantities, so predicting either of these is a **regression** problem, not classification.

Hence, the correct answers are: predicting whether a patient is diagnosed with a disease or not, and predicting whether it will rain or not tomorrow.

Consider the following confusion matrix for the classification of Hatchback and SUV:

		True	
		Hatchback	SUV
Prediction	Hatchback	55	5
	SUV	0	40

Question 3

Find the accuracy of the model.

- ☒ 0.95
- ☐ 0.55
- ☐ 0.45
- ☐ 0.88

Solution:

The accuracy of a classification model is the proportion of correctly classified observations among all observations:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}.$$

From the confusion matrix, the correct predictions are the diagonal elements: 55 and 40. Therefore,

$$\text{Correct predictions} = 55 + 40 = 95.$$

The total number of observations is

$$55 + 5 + 0 + 40 = 100.$$

Hence,

$$\text{Accuracy} = \frac{95}{100} = 0.95.$$

0.95

Question 4

Find the sensitivity of the model.

- ☐ 0.95
 - ☐ 0.55
 - ☒ 1
 - ☐ 0.88
-

Solution:

Sensitivity measures the ability of the classifier to correctly identify the positive class:

$$\text{Sensitivity} = \frac{TP}{TP + FN}.$$

Taking **Hatchback** as the positive class, $TP = 55$ because 55 actual Hatchbacks were correctly predicted as Hatchbacks. The false negatives are $FN = 0$, because no actual Hatchback was predicted as an SUV.

Therefore,

$$\text{Sensitivity} = \frac{55}{55 + 0} = \frac{55}{55} = 1.$$

Sensitivity = 1

Question 5

Under the ‘family’ parameter of `glm()` function, which one of the following distributions correspond to logistic regression for a variable with binary output?

- ☒ Binomial

- Gaussian
- Gamma
- Poisson

Solution:

Logistic regression is used when the dependent variable is categorical with two possible outcomes, e.g. $Y \in \{0, 1\}$.

A binary outcome can be modeled using the **binomial distribution**. In R, logistic regression is commonly fitted using

```
glm(..., family = binomial).
```

The binomial family is therefore used for logistic regression with binary output.

Binomial

Use the following information to answer Q6, Q7, Q8, Q9, and Q10:

Load the dataset `iris.csv` as a dataframe `irisdata`, with the first column as index headers, first row as column headers, dependent variable as factor variable, and answer the following questions.

The iris dataset contains four Sepal and Petal features (Sepal Length, Sepal Width, Petal Length, Petal Width, all in cm) of 50 equal samples of 3 different species of the iris flower (Setosa, Versicolor, and Virginica).

Question 6

What is the dimension of the dataframe?

- ✓ (150, 5)
- (150, 4)
- (50, 5)
- None of the above

Solution:

The Iris dataset contains 150 observations and 5 columns. The five columns consist of four flower measurements and one dependent/factor variable:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width
- Species

Therefore, the dimension of the dataframe is

Number of rows $\times$ Number of columns = 150×5 .

(150, 5)

Question 7

What can you comment on the distribution of the independent variables in the dataframe?

- ☐ The variables Sepal Length and Sepal Width are not normally distributed
 - ☒ All the variables are normally distributed
 - ☐ The variable Petal Length alone is normally distributed
 - ☐ None of the above
-

Solution:

The Iris dataset contains the following independent variables: Sepal Length, Sepal Width, Petal Length, and Petal Width.

Checking each variable's distribution (e.g. via a normality test or Q-Q plot) shows that all four variables are approximately normally distributed.

All the variables are normally distributed

Question 8

How many rows in the dataset contain missing values?

- ☐ 10
 - ☐ 5
 - ☐ 25
 - ☒ 0
-

Solution:

The Iris dataset used in the question contains 150 observations. Checking the dataset for missing values (e.g. using `sum(is.na(irisdata))` or similar) shows that there are no missing values in the dataset.

Therefore, the number of rows containing missing values is 0.

0

Question 9

Which of the following code blocks can be used to summarize the data (finding the mean of the columns `PetalLength` and `PetalWidth`), similar to the one given below?

PetalLength	PetalWidth
3.758000	1.199333

(Select all that apply.)

- ☒ `lapply(irisdata[, 3:4], mean)`
- ☐ `sapply(irisdata[, 3:4], 2, mean)`
- ☒ `apply(irisdata[, 3:4], 2, mean)`
- ☐ `apply(irisdata[, 3:4], 1, mean)`

Solution:

The relevant columns are columns 3 and 4 of the dataframe, selected using `irisdata[, 3:4]`.

`lapply()` applies the `mean` function to each selected column and returns the result as a list, with one mean value per column — this matches the required column-wise summary. `apply(irisdata[, 3:4], 2, mean)` applies `mean` across the second dimension (columns, since 2 denotes columns and 1 denotes rows in R's `apply()`), which also computes the mean of each column correctly.

On the other hand, `sapply(irisdata[, 3:4], 2, mean)` is invalid syntax — `sapply()` does not take a margin argument like `apply()` does. And `apply(irisdata[, 3:4], 1, mean)` computes the mean across rows instead of columns, which is not what is required. Therefore, both

`lapply(irisdata[, 3:4], mean)`

and

`apply(irisdata[, 3:4], 2, mean)`

are correct.

Question 10

What can be interpreted from the plot shown below?

The plot shows Sepal Width vs Sepal Length for the three Iris species: Setosa, Versicolor, and Virginica.

(Select all that apply.)

- ☐ Sepal widths of Versicolor flowers are lesser than 3 cm.
- ☒ Sepal lengths of Setosa flowers are lesser than 6 cm.
- ☐ Sepal lengths of Virginica flowers are greater than 6 cm.
- ☒ Sepals of Setosa flowers are relatively more wider than Versicolor flowers.

Solution:

Statement (Sepal length of Setosa): From the scatter plot, the Setosa observations are concentrated on the left side of the plot. Their Sepal Length values are generally below 6 cm.

Sepal lengths of Setosa flowers are lesser than 6 cm

is correct.

Statement (Sepal width comparison): The Setosa observations generally have larger Sepal Width values compared with Versicolor observations. Thus, Setosa flowers are relatively wider in terms of Sepal Width.

Sepals of Setosa flowers are relatively more wider than Versicolor flowers

is also correct.

The remaining two statements do not hold: Versicolor sepal widths are not consistently below 3 cm, and Virginica sepal lengths are not all greater than 6 cm (there is overlap with Versicolor).

Answer Summary

Q	Correct Answer
1	Bias variance trade off
2	Disease diagnosis; rain/no-rain prediction
3	0.95
4	1
5	Binomial
6	(150, 5)
7	All the variables are normally distributed
8	0
9	<code>lapply(...,mean)</code> and <code>apply(...,2,mean)</code>
10	Setosa: length < 6cm; wider sepals than Versicolor